

Nanthan Srikumar Radha

Backend & Infrastructure Engineer

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Authorized to work in Canada through June 2029, no sponsorship required • English (fluent), French (intermediate), Tamil

SKILLS

Languages | Python, C++, C, PHP, JavaScript, SQL, Bash / shell scripting
Backend & Data | FastAPI, REST API design, SSE, async programming, microservices, JWT auth, React, PostgreSQL, Apache Cassandra, SQLAlchemy, Alembic
Cloud & DevOps | AWS (ECS Fargate, ECR, S3, IAM OIDC), Oracle Cloud, Docker, Docker Compose, GitHub Actions CI/CD, Linux, Git, Nginx, Cloudflare Workers
Reliability | Distributed systems, high availability, failover, backup & recovery, production debugging, Prometheus, Grafana, pytest, unit & integration testing
ML & AI | MLflow, XGBoost, scikit-learn, Evidently, LLM integrations (Anthropic, Gemini, OpenRouter), Model Context Protocol (MCP), n8n, Claude Code multi-agent workflows

EXPERIENCE

Synopsys

Bangalore, India

R&D Engineer · Synopsys Common Licensing (SCL) Team

Jul 2023 – Jun 2024

- Migrated the production licensing datastore to **Apache Cassandra**; monitored cluster health and troubleshooted downtime across a **Python/C++/PHP** platform codebase to keep global client licensing available.
- Maintained **licensing servers across multiple zones** for enterprise client software distribution, validating **failover** paths and verifying **backup** integrity.
- Developed a **React** internal dashboard visualizing licensing and usage data, enabling leadership to track key metrics and identify suspected **software piracy**.

Research and Development Intern · Newton Data Model (NDM) Team

Jan 2023 – Jul 2023

- Investigated integration issues and production errors in the **Multi-Read/Write (M-ROW)** lock feature, analyzing **stack traces** to isolate recurring failure patterns and shipping targeted fixes across the codebase.
- Automated test-log triage in **Python**, parsing and aggregating execution logs to group related failures under parent categories and prioritize high-impact bugs, systematically clearing bulk test failures.

PROJECTS

MLOps Fraud Detection Pipeline | GitHub

- Built and deployed a **FastAPI** inference **microservice** containerized with **Docker** on **AWS ECS Fargate**, with a **GitHub Actions CI/CD** pipeline gating deploys on a passing **pytest** suite and authenticating to **ECR** keylessly via **IAM OIDC**.
- Wired the **MLflow model registry** @champion alias for **zero-downtime** promotion and **rollback**; instrumented **Prometheus** metrics with **Grafana** dashboards and **alert rules as code** over throughput, fraud rate, and **Evidently** data drift, plus asynchronous **S3** prediction logging for replay.
- Trained **XGBoost** on **284,807** transactions, optimizing **PR-AUC** under **577:1 class imbalance** and serving predictions in **2–5 ms**; **incident simulation** scripts exercise the full **alerting** path end-to-end.

Carnet Snap | Multi-tenant document-to-flashcard API

- Designed a **versioned REST API** in **FastAPI** with **JWT authentication**, per-user **quota metering**, and **Server-Sent Events** progress streaming over long-running jobs.
- Backed it with **Supabase PostgreSQL** under versioned **schema migrations** and an asynchronous **background worker pipeline**; **Gemini Vision** handles handwriting **OCR** across **9 languages**, with stable **MD5** IDs enforcing **idempotent** re-import.
- 242 pytest tests** run green with no API key, database, or external service, using **dependency injection** and fakes to keep the suite hermetic.

Self-Hosted Automation Infrastructure | Live

- Provisioned and operate an **Oracle Cloud ARM VM (4 OCPU, 24 GB)** running multiple containerized services under **Docker Compose**, fronted by an **nginx reverse proxy** with path-based routing, **DuckDNS** dynamic DNS, and automated Let's Encrypt **TLS** renewal.
- Built an **n8n** agency workflow framework with **five agentic phases** and per-client project structure; **four slash commands** automate deploy, maintain, monitor, and migrate operations.

EDUCATION

Concordia University

Master of Applied Computer Science

2024 – May 2026

Montreal, QC